Abdul Monaf Chowdhury

Website | LinkedIn | GitHub | Google Scholar

University of Dhaka, Bangladesh
Email: monafabdul15@gmail.com

Education

Master of Science in Robotics and Mechatronics Engineering August 2025 – Jan 2027
University of Dhaka, Bangladesh
Thesis: Value Propagation for Offline Goal Conditioned Reinforcement Learning
Supervisor: Dr. Md Mehedi Hasan

Bachelor of Science in Robotics and Mechatronics Engineering Jan 2019 – Jan 2024
University of Dhaka, Bangladesh
CGPA: ***3.87/4.00****, Ranked* 2nd ***place***

Relevant Coursework:
Artificial Intelligence, Introduction to Machine Learning, Digital Image Processing and Robot Vision, Digital Signal Processing, Human-Robot Interaction, Advanced Robotics

Skills:
Software: C, C++, Python, MATLAB, PyTorch, JAX, Flax, TensorFlow, LaTeX
Languages: English (fluent), Bangla (native)

Research Interests

Embodied AI, Reinforcement Learning, Multi-modal Learning, Computer Vision, Vision Language Models

Research Experience

Research Engineer Oct 2025 – Present
AVIS Lab, University of Dhaka *Dhaka, Bangladesh*
Supervisor: Dr. Md Mehedi Hasan

- Designed an efficient value learning algorithm extending the divide-and-conquer methods for offline goal-conditioned RL, and improved goal-reaching over the SOTA baselines on stochastic, deterministic, and stitching tasks in OGBench
- Evaluated goal-representation quality dependence on offline GCRL and developed Fourier positional encodings for state representation to improve downstream goal-reaching performance
- Two first-author papers are currently under review at **ICLR '27**

Research Assistant Feb 2024 – Sep 2025
MAIM Lab, University of Dhaka *Dhaka, Bangladesh*
Funding: Wellcome Leap (In Utero, California, USA)
PI: Dr. Niamh Nowlan, Co-PI: Dr. Abhishek Kumar Ghosh

- Collaborated with **University College Dublin** on the **Wellcome Leap In Utero** funded project titled "Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention"
- Designed and implemented deep learning-based frameworks to analyze multimodal sensor signals from wearable belts to detect body movements, fetal kicks, and fetal hiccups
- Optimized signal processing of sensor data, presented hardware design feedback based on analytical findings, and validated hardware design changes
- Assisted in attaining **$1M** funding extension, and eventually helped secure translational funding from Wellcome Leap to launch a **startup**

Professional Experience

Lecturer Mar 2026 – Apr 2026
Daffodil International University *Dhaka, Bangladesh*

- Instructed undergraduate junior and senior year students on Introduction to AI, and Control Systems courses, and took relevant labs.

Instructor May 2025 – Aug 2025
National Camp, Bangladesh AI Olympiad *Dhaka, Bangladesh*

- Instructed national camp students on Unsupervised Learning, Deep Learning, and Computer Vision algorithms and architectures, took relevant labs, and illustrated Deep Learning evaluation strategies and techniques

Publications

- *LaGEA: Language Guided Embodied Agents for Robotic Manipulation* **[Paper] [Code]**
  Abdul Monaf Chowdhury, AKM Moshiur Rahman Mazumder, Rabeya Akter Fariya, Safaeid Hossain
  43rd International Conference on Machine Learning, ICML '26

- *T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion* **[Paper] [Code]**
  Abdul Monaf Chowdhury, Rabeya Akter Fariya, Safaeid Hossain
  40th Annual AAAI Conference on Artificial Intelligence, AAAI '26

- *Beyond Visible Evidence: Open-World Amodal Counting Under Spatial Occlusion* **[Paper] [Code]**
  Safaeid Hossain, Rabeya Akter Fariya, **Abdul Monaf Chowdhury**, Md Mehedi Hasan, Md. Jubair Ahmed Sourov
  Winter Conference on Applications of Computer Vision - WACV 2027

- *U-ActionNet: Dual-Pathway Fourier Networks with Region-of-Interest Module for Efficient Action Recognition in UAV Surveillance* **[Paper]**
  Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan, Riad Ahmed, AKM Azad, Salem A. Alyami
  IEEE Access, 2024. Impact Factor = 3.4

- *FFT-UAVNet: FFT Based Human Action Recognition for Drone Surveillance System* **[Paper]**
  Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan
  5th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2023

Manuscript Submitted

- *GTRL: Grounding Divide-and-Conquer Value Learning with Temporal Differences* **[Paper] [Code]**
  Abdul Monaf Chowdhury, MD Sameer Iqbal Chowdhury, Shifat E. Arman, Md Mehedi Hasan
  International Conference on Learning Representations - ICLR 2027

- *Do Better Goal Representations Improve Goal-Conditioned Reinforcement Learning?* **[Paper] [Code]**
  Syed Nazmus Sakib*, **Abdul Monaf Chowdhury***, Nafiul Haque, Shifat E. Arman, Md Mehedi Hasan
  International Conference on Learning Representations - ICLR 2027
  * = *Equal-Contribution*

- *Contingency Planning with Latent World Models under Irreversible Outcomes*
  Abdul Monaf Chowdhury, Syed Nazmus Sakib, Ahanaf Ariq, Nafiul Haque, Shifat E. Arman, Md Mehedi Hasan
  Computer Vision and Pattern Recognition Conference - CVPR 2027

- *A Noise-Tolerant Wearable Device for Unsupervised In-Home Monitoring of Fetal Movements*
  Abhishek K. Ghosh, Saadman Sakib, **Abdul Monaf Chowdhury**, . . ., and Niamh Nowlan
  Information Fusion, 2026/27. Impact Factor = 17.4

- *Kernelized Parameter Sensitivity Atlases for Amortized Conditional Interpretability* **[Paper] [Code]**
  Davide Wiest, Ahanaf Hasan Ariq, **Abdul Monaf Chowdhury**
  Thirtieth Annual Conference on Artificial Intelligence and Statistics - AISTATS 2027

- *Certifying Whether to Adapt: Baseline-Relative PAC-Bayes–MMD Guarantees for Test-Time Adaptation* **[Paper]**
  Ahanaf Hasan Ariq, Hasaan Ahmad, Aayush Parekh, **Abdul Monaf Chowdhury**
  Thirtieth Annual Conference on Artificial Intelligence and Statistics - AISTATS 2027

Research Project

Offline Goal-Conditioned Reinforcement Learning [Code] [Paper] Jan 2026 – Sep 2026
Reinforcement Learning, Representation Learning

- Designed an efficient value learning algorithm extending the divide-and-conquer methods for offline goal-conditioned RL, and improved goal-reaching over the SOTA baselines on stochastic, deterministic, and stitching tasks in OGBench
- Evaluated goal-representation quality dependence on offline GCRL and developed Fourier positional encodings for state representation to improve downstream goal-reaching performance

Open World Amodal Counting [Code] [Paper] Jan 2026 – Apr 2026

Multimodal Learning, Vision Language Model, Representation Learning

- Architected a hierarchical Feature Reconstruction Module fusing visible context with vision–language priors to reconstruct occluded class-discriminative features, anchored by teacher–student Visual-Equivalence consistency
- Built occlusion-augmented FSC-147 and CARPK benchmarks; achieved SOTA amodal counting from RGB alone, reducing MAE by **32.2%** (FSC-147), **40.7%** (CARPK), and **28.52%** (CAPTURe-Real, cross-dataset)

Language Guided Embodied Agents [Code] [Paper] Aug 2025 – Jan 2026

Embodied AI, Vision Language Model, Reinforcement Learning

- Designed a **Qwen2.5-VL-3B**-driven "episodic self-reflection" module that generates natural-language self-assessments of each trial, pinpointing successes and failure causes
- Fused self-reflections with CLIP-style vision–language feedback and goal images into a dense, semantically grounded reward for **Soft Actor-Critic**, accelerating convergence on Meta-World manipulation tasks

Tri-Modal Time Series Forecasting [Code] [Paper] Apr 2025 – Aug 2025

Large Language Model, Deep Learning, Signal Processing

- Architected an Adaptive Dynamic Multi-Head Cross-Modal Attention module with channel-wise residual skip-connections for fine-grained alignment of temporal and auxiliary features
- Engineered an **FFT**-based frequency-domain encoder and a Rich-Horizon Gating Fusion of spectral and temporal encodings; beat **SOTA** on multivariate time-series forecasting

Human Action Recognition from Drone footage [Paper] Jan 2023 – Jan 2024

Computer Vision, Deep Learning

- Designed a hybrid model combining a modified 3D CNN with an FFT-based action recognition module to detect suspicious human activity in UAV surveillance
- Built a lightweight MobileNetV2 + BiLSTM pipeline human activity detection for edge devices, reducing inference time

Proxemics & Social Interaction Patterns in ASD Children Sep 2023 – Jan 2024

Human-Robot Interaction, Deep Learning

- Formulated a **YOLOv8**-based system to determine the ideal proxemics of children with autism spectrum disorder (ASD) interacting with a **NAO** robot
- Analyzed the behavioural responses of 20 children with ASD to specific NAO robot actions

Awards & Scholarships

- Dean's Award for Best Undergraduate Result, University of Dhaka, 2024
- Engineering Faculty Undergraduate Merit Scholarship, University of Dhaka, 2024
- 5th, Dataverse Challenge - ITVerse, Bangladesh, 2023; [Report]
- 2nd, Intra-Department Soccer Bot Championship, University of Dhaka, 2019
- Sylhet Board Scholarship, Higher Secondary Certificate Examination 2018

Workshop/Conference Attended

- 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka
- 40th Annual AAAI Conference on Artificial Intelligence, AAAI '26
- 43rd International Conference on Machine Learning, ICML '26

Reviewer

- AAAI 27 **Program Committee**, AAAI 26, ICML 26, ICLR 27, IEEE Access

Leadership/Volunteer Activities

General Secretary Mar 2022 – Feb 2024
RMEDU Student Club, University of Dhaka

Academic Team Mentor Sep 2019 – Aug 2022
Bangladesh Robot Olympiad

Program Co-Ordinator Jul 2021 – Jun 2022
IEEE Robotics & Automation Society, University of Dhaka